

SiGe Photodetector Using a Tapered Ge Design for 400 Gbps Optical Links

Santiago Bernal^{1,*}, Deng Mao², Charles Lin², Charles St-Arnault¹, Aleksandar Nikic¹,
Benton Qiu¹, Kaibo Zhang¹, Yannick D'Mello¹, Masaki Kato², David V. Plant¹

¹Dept. of Electrical and Computer Engineering, McGill University, Montreal QC H3A 0G4, Canada

²Marvell, Santa Clara, CA 95054 USA

*Santiago.Bernal@mail.mcgill.ca

Abstract: We report the DC, small- and large-signal performance of a CMOS-compatible tapered-germanium photodetector with 0.87 A/W responsivity and up to 400 Gbps PAM-8 operation, enabling high-speed silicon photonic interconnects. © 2026 The Author(s)

1. Introduction

Data center growth and AI have spurred the need for faster, cheaper, and more sustainable optical components. As modulators, digital to analog converters (DACs), and laser technologies continue to push transceiver speed boundaries [1-3], the need for mass producible receivers with matching specifications emerges. Currently, silicon photonics (SiP) is a leading solution platform due to its growing maturity and compatibility with complementary metal-oxide-semiconductor (CMOS) processes. This compatibility helps reduce costs and increase yields of building photonic integrated circuits (PICs). As the SiP platform advances towards large-scale integration, the demand for high performance, CMOS-compatible components, specifically Germanium-on-Silicon photodetectors (PDs), has significantly increased [4].

SiP PDs have been optimized to have high bandwidths, responsivity, and low noise [5]. The bandwidth of these PDs is determined by the transit time of the absorbed electrons and equivalent RC circuits. However, high capacitances impede achieving bandwidths over 110 GHz using standard CMOS fabrication methods. To reduce this limitation, Ge PDs have been fabricated using smaller footprints which reduces the responsivity since the optical mode absorption is inversely proportional to the device length. To increase responsivity, PDs can be fabricated using various methods such as sharp corner reflectors [6] or distributed Bragg reflectors [7] to increase absorption via additional backward propagation. Recent works have demonstrated compact PDs with bandwidths on the order of 100 GHz and responsivities near 1 A/W in SiP platforms [8]. These methods help lower the penalties from smaller footprints and can often be combined with bandwidth improving techniques. While the higher capacitance limitation can be mitigated by using an inductive-gain-peaking effect, this approach is undesirable. It restricts the co-designing of transimpedance amplifiers and other packaging requirements. Alternatively, in-situ metal depositions can also be used to reduce the capacitance and shorten transit times of Ge PDs to achieve bandwidths of over 265 GHz [9]. However, these techniques are not available to most CMOS foundries and therefore cannot meet the mass production required for scaling.

In this paper, we report a novel tapered-Ge PD built using a standard SiP foundry process capable of net data rate of 400 Gbps, with 71 nA dark current at -1 V and 0.87 A/W responsivity in the O-band. We introduce a new method to increase responsivity by enhancing optical mode confinement with a Ge taper, ensuring only the fundamental mode couples to the Ge layer. We further evaluate the PD response in an intensity-modulated direct-detection (IMDD) system with different symbol rates and received optical powers. We also present the resulting eye diagrams.

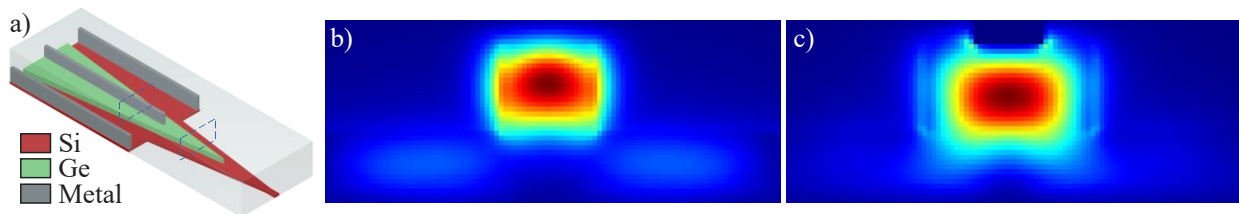


Fig. 1: (a) Simplified 3D schematic of the tapered-Ge PD with cross section locations (blue dashed). (b) Simulated electric field intensity cross section at the input of the tapered-Ge. (c) Simulated electric field intensity cross section inside the PD junction.

2. Design

Fig. 1a is a simplified three-dimensional schematic of the Si tapered-Ge PD. It consists of a vertical $p-i-n$ junction structure with a single top metal contact. Light couples into the PIC containing the PD via an edge coupler followed by a tapered Si waveguide. The Ge layer is also tapered ensuring only the fundamental optical mode couples from Si into Ge. Fig. 1b and 1c show the simulated cross-sectional electric field intensity (E) using Ansys Lumerical MODE, at the start of the Ge taper and at the junction, respectively. The tapered Ge structure allows for an increase of E inside the Ge layer since only single mode propagation is allowed which increases the overall absorbed light. Coupling was also optimized using finite-difference time domain (FDTD) simulations. Coupling efficiency between the two materials was increased by the gradual taper of the Si waveguide before the Ge region. The tapered-Ge design exhibited an 18% increase in simulated optical power absorption compared to a bulk-Ge region. This increase directly leads to an improvement in responsivity. The bandwidth of the PD was optimized using Ansys Lumerical CHARGE simulations considering both the RC constant of the junction and the carrier transit time.

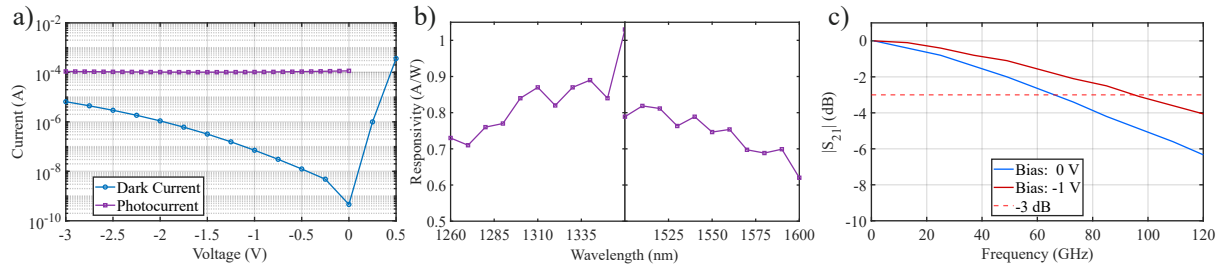


Fig. 2: Small signal characterization of the PD. (a) Measured dark current and photocurrent at -8 dBm power with varying bias voltages. (b) Measured responsivity at different wavelengths in the O and C-bands. (c) Simulated O/E S21 for 0V and -1V bias.

3. Results

The PIC was fabricated using a standard SiP foundry process. The dark current, photocurrent, and responsivity measurements for -8 dBm optical power into the PD are seen in Fig. 2a and 2b. A DC bias was applied to the PD using a 110 GHz probe and bias tee. The current was measured using a pico-ammeter and the bias voltage was swept from -3 V to 0.5 V. The dark current at -1 V and -3 V is 71 nA and 6.5 μ A, respectively. We believe the reason for the high dark currents at higher voltages is due to the proximity of the anode and cathodes contacts which generate a strong electric field in the junction. The flat photocurrent response indicates that the strong electric field did not trigger impact ionization. This process is typical in avalanche photodiodes, which also exhibit increased dark current as the bias approaches breakdown [7]. The responsivity was then measured across both telecommunication bands. The intrinsic responsivity was 0.87 A/W at 1310 nm, with a maximum of 1.03 A/W in the O-band. In the C-band, the maximum value was 0.81 A/W with 0.74 A/W at 1550 nm. The O/E S21 was simulated to be 62 GHz and 95 GHz at 0 V and -1 V bias as shown in Fig. 2c, respectively. Importantly, the device has high bandwidths without using inductive peaking, making it more suitable for next-generation data transmissions.

Fig. 3a shows the experimental setup used to characterize the large signal performance of the PD in an IMDD system. At the transmitter, an O-band 13 dBm tunable laser is input into a polarization controller (PC) and then into a fully packaged, 110 GHz Hyperlight thin-film lithium niobate (TFLN) Mach-Zehnder modulator (MZM). We use a Keysight M8199B arbitrary waveform generator (AWG) with a 256 GSa/s DAC channel to directly modulate the MZM for various amplitude modulation formats. At the receiver, we use a Keysight UXR, 256 GSa/s real-time oscilloscope (RTO) to recover our signal. The PD is biased using a source measuring unit (SMU) and a bias tee with its output directly connected to the oscilloscope, without any electrical amplifier. The received waveforms are then resampled to two samples per symbol and synchronized. Finally, we apply equalization, and the symbols are mapped back into bits to calculate the bit error rate (BER). The resulting eye diagrams are also generated.

The IMDD transmission results and resulting eye diagrams for PAM-2,4,8 can be seen in Fig. 3b-3d. For each modulation format, the symbol rate was swept to achieve the highest net throughput after forward error correction (FEC). For PAM-2, the lowest pre-FEC BER achieved was 10^{-6} , which is the lowest measurable BER in our setup. This corresponds to error-free transmission and was possible until 160 Gbaud. For PAM-2, a net rate of 160 Gbps was obtainable under the KP4-FEC at 170 Gbaud and a maximum net rate of 178 Gbps was achieved under the hard-decision (HD) FEC at 190 Gbaud. For PAM-4, the highest throughputs under the HD- and the O-FECs were at 160 and 170 Gbaud for a net 300 Gbps and 298 Gbps, respectively. For PAM-8, net 400 Gbps was achieved under the

25% overhead soft-decision (SD) FEC threshold with 170 Gbaud. The generated eye diagrams for PAM-2,4,8 are shown at the top, middle, and bottom of Fig. 3d, respectively. The PD sensitivity was also tested by varying the input power into the device. The highest received optical power (ROP) was 5 dBm. Fig. 3c shows the BER at 170 Gbaud for PAM-2,4,8 with different ROPs. PAM-2 maintained its FEC threshold with up to a 1.4 dB reduction in ROP, while PAM-4 tolerated a 1.5 dB reduction. In contrast, PAM-8 showed only 0.2 dB tolerance. This is due to the strict signal-to-noise requirements of higher modulation formats which lead to higher inter symbol interference at low ROPs and require stronger digital signal processing techniques to compensate, especially at high symbol rates [1, 2]. These results highlight the potential of SiGe PDs fabricated using standard SiP foundry processes to meet requirements of high-speed data transmission systems.

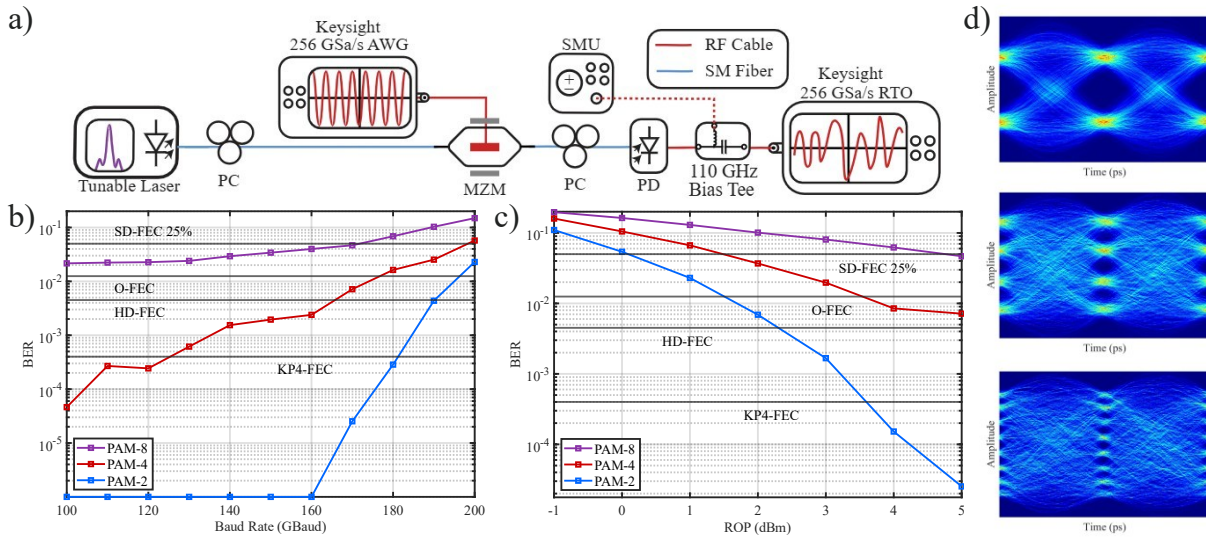


Fig. 3: (a) Experimental setup for small and large system measurements. (b) BER versus symbol rate for PAM-2,4,8. (c) BER versus received optical power at 170 Gbaud. (d) Eye diagrams at 170 Gbaud PAM-2 (top), PAM-4 (middle), PAM-8 (bottom).

4. Conclusion

We simulated, fabricated and measured a novel Germanium-on-Silicon vertical PD design compatible with CMOS technology. We present the DC, small signal, and large signal performance of a tapered-Ge PD. It achieved a dark current of 71 nA at -1V with a responsivity of 0.87 A/W at 1310 nm and 0.74 A/W at 1550 nm. Net throughputs of 178 Gbps, 300 Gbps, and 400 Gbps were demonstrated for PAM-2,4,8 under HD-FEC, O-FEC, and 25% SD-FEC, respectively. The PD also exhibited 1.5 dB ROP tolerance for PAM-2 and PAM-4 at high symbol rates demonstrating its potential suitability for high-bandwidth optical interconnects.

Acknowledgements

The authors would like to thank Testforce Systems and Rohde & Schwarz.

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